

```
> install.packages("skimr")
Installing package into '/home/shakthi/R/x86_64-pc-linux-gnu-
library/4.1'
...

* testing if installed package can be loaded from temporary
location
* testing if installed package can be loaded from final
location
* testing if installed package keeps a record of temporary
installation path
DONE (skimr)

> library(skimr)
```

The *skim()* function gives a broad overview of the data frame. An example for the training data is given below:

```
> skimData <- skim(train)

> skimData
— Data Summary —————

Name                Values
Number of rows      150
Number of columns    1

Column type frequency:
  numeric            1

Group variables      None

— Variable type: numeric —————
  skim_variable n_missing complete_rate mean   sd p0  p25
p50  p75 p100 hist
1 data          0           1 105. 60.7  1 53.2
106. 154.  214 00000
```

downSample()

You can use the *downSample()* function to randomly obtain data from a set such that all the classes have the same frequency as the minor class. An example using the oil data set is illustrated below:

```
> data(oil)

> table(oilType)
oilType
 A  B  C  D  E  F  G
37 26  3  7 11 10  2

> downSample(fattyAcids, oilType)
```

```
Palmitic Stearic Oleic Linoleic Linolenic Eicosanoic
Eicosenoic Class
1   9.8  5.3 31.7  51.3   0.8  0.4  0.2   A
2  11.0  5.3 35.0  45.2   1.3  1.3  0.7   A
3   5.9  4.5 24.1  61.7   0.9  0.6  0.6   B
4   6.2  4.0 28.3  59.7   0.9  0.1  0.1   B
...
```

The *downSample()* function accepts the following arguments:

Argument	Description
directory	A character vector of file path names
x	A matrix or data frame of variables
y	A factor variable
list	Boolean value to return a list or not
yname	A label for the class column

upSample()

The *upSample()* function is similar to the *downSample()* function for the fatty acids data where additional samples are added to the minor class as shown below:

```
> upSample(fattyAcids, oilType)
Palmitic Stearic Oleic Linoleic Linolenic Eicosanoic
Eicosenoic Class
1   9.7  5.2 31.0   52.7   0.4  0.4  0.1   A
2  11.1  5.0 32.9   49.8   0.3  0.4  0.1   A
3  11.5  5.2 35.0   47.2   0.2  0.4  0.1   A
4  10.0  4.8 30.4   53.5   0.3  0.4  0.1   A
...
```

You are encouraged to read the *caret*, *mda* and *mlbench* package reference manuals to learn more functions, arguments and their usage. 

References

- *caret* reference; <https://cran.r-project.org/web/packages/caret/caret.pdf>
- *mda* reference; <https://cran.r-project.org/web/packages/mda/mda.pdf>
- *mlbench* package; <https://cran.r-project.org/web/packages/mlbench/mlbench.pdf>

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